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CSC 270

16 September 2026

Pseudo Code for Assignment 3

Non-Negotiatables:

First when writing a computer program in C++, we write the 4 lines of code.

```
#include <iostream>
#include <cmath>

using namespace std;

int main(){

}
```

Defining Variables:

After writing these lines, I started to solve the problem by defining necessary variables.

```
double t, v;
bool ns = true;
```

The “t” represents the time(in minutes) when you input how long it took the driver to get from CSI to JFK. You will have to give input when it's asked.

```
cout << "How long did it take for the car to get to JFK in minutes?: ";
cin >> t;
```

The “v” represents the velocity that's calculated by dividing the distance which is 27(given) and the time(what the person has to input.)

```
v = 27/t;
```

In order for me to keep this in a while loop, I used a boolean called “ns”(short for not safe). I kept this as true so that the loop will keep on running and ask for input until “ns” turns into false(which is when the time it took is safe).

```
ns = false;
```

The Main Program:

Once you input a number when asked, the program will run and determine two things. The first thing it looks at is if the time it took to get to JFK is a legal speed(26 and 53 which is given), then it will do the equation to solve for the velocity and determine if the velocity is safe (0.509434 < v < 1.03846). If the time is illegal and the velocity isn't in between these 2 numbers, the process will repeat until you get a legal time and a velocity that is deemed safe.

```
    while(ns == true)
        if(t < 26 && v > 1.03846){
        cout << "You were going too fast! You are getting a $200
ticket!\n";
        cout << "Velocity: ";
        cout << v ;
        cout << " m/s\n";
        cout <<
"--------------------------------------------------------------------------
---------------------------------------\n";
        cout <<
"--------------------------------------------------------------------------
---------------------------------------\n";
        cout << "How long did it take for the car to get to JFK in
minutes?: ";
        cin >> t;
```

```
        cout <<
"------------------------------------------------------------------------
---------------------------------------\n";
        cout <<
"------------------------------------------------------------------------
---------------------------------------\n";
        v = 27/t;
        }

        else if(t > 53 && v < 0.509434){
        cout << "You are going too slow! You are getting a $50 ticket!\n";
        cout << "Velocity: ";
        cout << v ;
        cout << " m/s\n";
        cout <<
"------------------------------------------------------------------------
---------------------------------------\n";
        cout <<
"------------------------------------------------------------------------
---------------------------------------\n";
        cout << "How long did it take for the car to get to JFK in
minutes?: ";
        cin >> t;
        cout <<
"------------------------------------------------------------------------
---------------------------------------\n";
        cout <<
"------------------------------------------------------------------------
---------------------------------------\n";
        v = 27/t;
        }

        else{
        cout << "You are safe!\n";
        cout << "Velocity: ";
        cout << v ;
        cout << " m/s";
        ns = false;
        }
}
```

—----------

Other Things To Note:

There are some things that I wrote that we did not discuss in class. Using my knowledge in python, I noticed that these things actually work also in C++. These things though are mostly used for formatting.

For example, I used the “\n” feature in this program to make the program go to a new line rather than having everything cramped into one line.

```
cout << " m/s\n";
        cout <<
"--------------------------------------------------------------------------
--------------------------------------\n";
        cout <<
"--------------------------------------------------------------------------
--------------------------------------\n";
```

The lines also as seen above that I print are used to organize what is seen in the terminal, keeping them from being cramped.

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VS<V<VF:

Given the safe travel time window: 26 < t < 53 (minutes)
Velocity formula: V = 27 / t

Since velocity is inversely proportional to time, the bounds flip:

t = 26 minutes --> V = 27/26 = 1.03846 m/s (VF)
t = 53 minutes --> V = 27/53 = 0.509434 m/s (VS)

Answer:

VS < V < VF

VS = 27/53 = 0.509434 m/s
VF = 27/26 = 1.03846 m/s

So the safe velocity range is approximately:

$0.509434 \text{ m/s} < V < 1.03846 \text{ m/s}$